Sign Language Recognition using Temporal Classification

Mini-Project Write-up | Recognition across 95 sign classes

1. Problem Statement

Sign-language recognition maps recorded movements or sensor measurements to a sign class. This project classifies Auslan sign recordings represented as time-series features, predicting one of 95 labels. Because recordings differ in duration and quality, the project evaluates high-quality and low-quality feature representations and compares traditional machine-learning classifiers with neural-network approaches.

2. Dataset Details

The high-quality representation contains 2,565 recordings across 95 classes, represented as 136 time steps with 22 features per step. The low-quality set contains 6,648 valid recordings across the same 95 classes; the selected representation has 8 features and was resampled to 100 time steps. One empty low-quality recording was skipped. The high-quality CSV representation was obtained from the MFDS GitHub repository, while low-quality recordings came from the Auslan .sign archive. These are prepared data representations and should not be described as an exact reproduction of every raw-data processing step in the reference paper.

Split strategy: Both datasets were split using a stratified 80:20 recording-level split with random seed 42; all 95 classes occur in both subsets. The high-quality split contains 2,052 training and 513 test recordings. The low-quality split contains 5,318 training and 1,330 test recordings.

3. Approach and Implementation

Preprocessing included selecting relevant low-quality sensor features, skipping the empty recording, resampling variable-length sequences to fixed length, and saving train/test arrays. Neural-network features were normalized using training-set statistics. Traditional classifiers used flattened sequence vectors with a StandardScaler fitted on training data only.

Logistic Regression and an RBF-kernel Support Vector Machine (SVM) were evaluated on both datasets. A feed-forward Multilayer Perceptron (MLP) was additionally trained on the high-quality data. The MLP uses 2,992 flattened input features, Dense layers with 256 and 128 ReLU units, dropout regularization, and a 95-class softmax output. LSTM variants were also explored for temporal classification, but their validation performance remained poor and they were not the best-performing approach.

4. Experimental Results

Dataset	Model	Accuracy	Macro F1
High-quality	**MLP**	**96.49%**	96.55%
High-quality	Logistic Regression	95.71%	95.67%
High-quality	SVM (RBF)	74.07%	74.91%
Low-quality	SVM (RBF)	64.14%	63.91%
Low-quality	Logistic Regression	46.92%	46.38%

The MLP achieved the highest measured test accuracy on the high-quality dataset (96.49%), narrowly exceeding Logistic Regression (95.71%). On low-quality data, SVM outperformed Logistic Regression. The MLP's macro precision, recall, and F1-score were 97.19%, 96.63%, and 96.55%. These are results from a held-out recording-level test split; signer-independent evaluation and statistical significance testing were not performed.

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Implementation, discussion, and conclusion

5. Evaluation and Demonstration

Evaluation used accuracy, macro precision, macro recall, and macro F1-score. A classification report and confusion matrix were generated for the MLP across all 95 classes. It achieved 96.49% accuracy on 513 high-quality test recordings. The interactive notebook demo lets a user select an existing held-out recording and displays the actual label, predicted label, and whether they match. This is a dataset-based demo, not a live webcam recognizer or a system that accepts arbitrary new videos.

6. Relation to the Research Paper

The reference work, “Sign Language Recognition using Temporal Classification,” discusses Logistic Regression and SVM baselines and explores LSTM and Sequential Pattern Mining (SPM) approaches for lower-quality data. This project implements Logistic Regression and SVM baselines, explores LSTM variants, and adds an MLP as an additional classifier. The results are experiments on the prepared dataset representations described here, not an exact reproduction of every experiment in the paper. The MLP is a feed-forward classifier over flattened features; unlike an LSTM, it does not explicitly model temporal dependencies.

7. Limitations

- The current demo uses pre-recorded labelled test examples and does not perform live camera-based recognition.
- The high-quality split is recording-level, not signer-independent; generalization to unseen signers has not been established.
- The MLP was evaluated only on the high-quality dataset, so its performance cannot be compared across both dataset qualities.
- The LSTM experiments did not achieve useful classification performance in the current configuration.
- No statistical significance test or independent external test set was used.

8. Conclusion

The project established a preprocessing and evaluation workflow for 95-class sign recognition using high-quality and low-quality Auslan feature representations. The MLP performed best on the high-quality held-out test set with 96.49% accuracy, while SVM was the stronger traditional baseline on the low-quality dataset. The tested LSTM setups did not perform well under the current training configuration. Future work should evaluate signer-independent splits, improve sequence-aware approaches, test the MLP on low-quality data, and build a live-input pipeline if real-time recognition is required.

References

[1] H. Cate, F. Dalvi, and Z. Hussain, “Sign Language Recognition using Temporal Classification,” project report (2015); arXiv:1701.01875. https://arxiv.org/abs/1701.01875
[2] MFDS repository used for the high-quality CSV representation: https://github.com/Halmaris/mfds
[3] Auslan .sign archive used for the low-quality recordings (UCI repository archive).